

Tianhao Fei

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Personal Information

- **Date of Birth:** March 6, 2002
- **Marital Status:** Single
- **Gender:** Male
- **Home:** Anyue County, Ziyang City, Sichuan Province, China

Education

Master, University of Science and Technology Beijing

September 2024 – June 2027

School of Automation and Electrical Engineering, Control Engineering

- Score: 90.1/100; GPA: 4.0/4.0.
- Supervisor: **Yongliang Yang**

Bachelor, University of Science and Technology Beijing

September 2020 – June 2024

School of Automation and Electrical Engineering, Automation

Research Interests

Robotics, Reinforcement Learning, Adaptive Dynamic Programming, Optimal Control, Game Theory, Autonomous Driving, Unmanned Systems.

Research Experience

1. Adaptive Backstepping Control for Constrained Uncertain Nonlinear Systems

- **Adaptive control under multiple uncertainties:** Developed backstepping-based neural control schemes for nonlinear constrained systems with unknown parameters, stochastic disturbances, and unknown control directions.
- **Distributed synchronization and robust tracking:** Designed command-filtered and compensation-based controllers for heterogeneous switched multi-agent systems and time-delay dynamics under finite-energy disturbances.
- **Stability and performance guarantees:** Established boundedness, constraint satisfaction, bipartite synchronization, and robust H_∞ tracking performance through Lyapunov-type, and zero-sum game formulations.

2. Projected Self-Triggered Reinforcement Learning for Differential Games

- **Self-triggered learning:** Proposed communication-efficient learning frameworks for zero-sum and nonzero-sum differential games, avoiding continuous state monitoring while preserving closed-loop stability.
- **Projected safe adaptation:** Incorporated projection operators into Actor–Critic and Actor–Critic–Disturbance learning laws to prevent parameter drift and keep adaptive weights within prescribed compact sets.
- **Resource-aware Nash approximation:** Combined intermittent feedback, fuzzy logic, and experience replay to approximate Nash policies with reduced sampling, bounded learning errors, and Zeno-free execution.

3. Data-Efficient Composite Actor–Critic Reinforcement Learning

- **Composite learning for game-theoretic problem:** Developed Actor–Critic learning algorithm for multi-player nonzero-sum differential games by integrating instantaneous Bellman errors with historical data.
- **Relaxed excitation requirements:** Replaced restrictive persistence of excitation condition with finite/sufficient excitation-based learning, improving the practicality of online adaptive dynamic programming.
- **Improved convergence and stability:** Established convergence of critic/actor weights and closed-loop stability while enabling approximate Nash equilibrium learning for nonlinear multi-agent decision systems.

Publications

Journal:

1. T. Fei, *et. al*, “[Game-Theoretic Fuzzy Reinforcement Learning via Self-Triggered Mechanism](#)”. IEEE Transactions on Fuzzy Systems, early access, 2026, doi: 10.1109/TFUZZ.2026.3706062. **(JCR Q1)**
2. T. Fei, *et. al*, “[Self-Triggered \$H_\infty\$ Control Using Projected Actor-Critic-Disturbance Learning](#)”. IEEE Transactions on Emerging Topics In Computational Intelligence, early access, 2026, doi: 10.1109/TETCI.2026.3683743. **(JCR Q1)**
3. T. Fei, *et. al*, “[Adaptive Neural Design for Permanent Magnet Synchronous Motor with Asymmetric Constraints and Input Dead-Zone](#),” Neurocomputing, vol. 640, p. 130294. **(JCR Q1)**
4. T. Fei, *et. al*, “[Command Filtered Adaptive Bipartite Synchronization for Heterogeneous Switched Multi-Agent Systems with State Delays and Input Nonlinearity](#),” Journal of the Franklin Institute, vol. 363, no. 2, p. 108395. **(JCR Q1)**
5. T. Fei, *et. al*, “[Robust Adaptive \$H_\infty\$ Tracking Control for Nonlinear Strict-Feedback System via Actor-Critic-Disturbance Learning: A Zero-Sum Game-Based Approach](#),” International Journal of Robust Nonlinear Control, vol. 36, no. 8, pp. 4517-4536. **(JCR Q1)**

Conference:

7. T. Fei, *et. al*, “[Self-triggered Sampling Design For Continuous-Time Linear Dynamical System](#),” 2025 China Automation Congress (CAC), Harbin, China, 2025, pp. 3164-3169.
8. T. Fei, Y. Yang*, “Self-Triggered Nash Equilibrium Seeking for Multi-Player Linear Quadratic Differential Games,” 2026 Chinese Control and Decision Conference (CCDC), Nanjing, China, 2026. **Accepted, Oral presentation.**
9. T. Fei, *et. al*, “Communication-Efficient Reinforcement Learning for Linear Quadratic Differential Zero-Sum Games,” 2026 International Annual Conference on Complex Systems and Intelligent Science (CSIS-IAC), Hefei, China, 2026. **Accepted, Best Paper Award.**

Patent:

10. Y. Yang, T. Fei, *et. al*, “[Self-Triggered Robust Reinforcement Learning Method and Apparatus for Zero-Sum Games of Unmanned Surface Vehicles](#)”. Publication No. CN122151484A.
11. Y. Yang, T. Fei, *et. al*. “[Resilient Control for Power Buffers in DC Microgrids Against Denial-of-Service Attacks](#)”. Publication No. CN121566407A.
12. Y. Yang, Z. Shan, T. Fei, *et al*. “[Composite Game-Theoretic Reinforcement Learning for Quadrotor UAV Swarms](#)”. Publication No. CN121879395B.
13. Y. Yang, Z. Shan, T. Fei, Q. Li. “[Secure Control Method and Device for Power Buffers in DC Microgrids Against FDI Attacks](#)”. Publication No. CN121440520A.

Honors and Awards

- **2026 China Moutai “Pillars of the Nation” Elite Talent Award**, [China Youth Development Foundation](#) & [Kweichow Moutai Group](#). (150 recipients nationwide)
- **2026 Best Paper Award**, 2026 International Annual Conference on Complex Systems and Intelligent Science.
- **2026 Commemorative Certificate for Research Achievement in Invention Patent**, “Guohong Zhongjing Cup”, School of Automation and Electrical Engineering, University of Science and Technology Beijing (USTB).
- **2025 Excellent Merit Master Student**, USTB. (TOP 5%)
- **2025 Inspur “Langchao” Enterprise Scholarship**, [Inspur Group Co., Ltd.](#) (16 recipients university-wide)
- **2025 First-Class Academic Scholarship for Master Students**, USTB. (TOP 7%)
- **2023 Second Prize, Mathematical Competition** of USTB.
- **2022 Social Practice Gold Award**, USTB.

Skills

Languages Mandarin Chinese (Native), **CET-6**.

Coding Matlab, Python, $\text{\LaTeX}$...

Misc. Academic research, teaching, training, consultation, $\text{\LaTeX}$ typesetting and publishing.